Tutorial - 09 KNN (cross valid

Q: A retail company wants to predict customer purchase behavior using KNN and evaluate it using k-fold cross-validation.

Input Features:

- Age
- Annual Income (in thousands)
- Shopping Frequency (visits per month)
- Loyalty points.

Target variable:

- Purchase (Yes = Y, No = N)

Complete Dataset (12 samples):

Age	Annual Income	Shopping Frequency	Purchase
25	50	8	Y
45	80	3	N
28	55	9	Y
50	85	2	N
30	60	10	Y
48	82	3	N
26	52	8	Y
47	88	2	N
29	58	9	Y
46	84	3	N
27	54	8	Y
49	86	2	N

Questions :-

① Divide the dataset into 5 folds for k-fold cross validation (4 samples per fold). List the customer IDs in each fold ensuring balanced class distribution.

② Perform 5-fold as validation for KNN with $k=3$

Training samples (12 customers)

validation samples (4 customers)

③ complete the cross-validation for all 5 folds and create a summary table:

20 datasets

(20) -> 5 fold k-fold CV.

then $\frac{20}{5} = 4$ data points in each fold

20
Train (16) validation (4)

Fold - 1 (16, 4) 16 - Training
4 -> validation

4	4	4	4	4

Train Test

=> $k=3$ as mentioned in question.

(k = 5)

① Distance Finding

Fold - 1 (9, 3)

Distance for (46, 84, 3) to (25, 50, 8)
(45, 80, 3)
(28, 55, 9)
(50, 85, 2)
(30, 60, 10)
(48, 82, 3)
(26, 52, 8)
(47, 88, 2)
(29, 58, 9)

① (46, 84, 3) to (25, 50, 8)

$\Rightarrow \sqrt{(46-25)^2 + (84-50)^2 + (3-8)^2}$

$= \sqrt{}$

$= \sqrt{} = 40.29$

② (46, 84, 3) to (45, 80, 3)

$\Rightarrow \sqrt{(46-45)^2 + (84-80)^2 + (3-3)^2}$

$= 4.123$

③ (46, 84, 3) to (28, 55, 9)

$\Rightarrow \sqrt{(46-28)^2 + (84-55)^2 + (3-9)^2}$

$= 34.65$

④ (46, 84, 3) to (50, 85, 2)

$\Rightarrow \sqrt{(46-50)^2 + (84-85)^2 + (3-2)^2}$

$= 4.24$

⑤ (46, 84, 3) to (30, 60, 10)

$\Rightarrow \sqrt{(46-30)^2 + (84-60)^2 + (3-10)^2}$

$= 29.68$

(6) (46, 84, 3) to (48, 82, 3)

$= \sqrt{(46-48)^2 + (84-82)^2 + (3-3)^2}$

$= 2.82$

(7) (46, 84, 3) to (26, 52, 8)

$= \sqrt{(46-26)^2 + (84-52)^2 + (3-8)^2}$

$= 30.06$

(8) (46, 84, 3) to (47, 88, 2)

$= \sqrt{(46-47)^2 + (84-88)^2 + (3-2)^2}$

$= 4.24$

(9) (46, 84, 3) to (29, 58, 9)

$= \sqrt{(46-29)^2 + (84-58)^2 + (3-9)^2}$

$= 31.6$

9 training samples $\boxed{k=5}$

40.29

4.123 N

34.65

4.24 N

29.68 Y

2.82 N

30.06

4.24 N

31.6

Distance for (27, 54, 8) to (25, 50, 8)

(45, 80, 3)
(28, 55, 9)
(50, 85, 2
(30, 60, 10
(48, 82, 3
(26, 52, 8
(47, 88, 2
(29, 58,

① (27, 54, 8) to (25, 50, 8)

$= \sqrt{(27-25)^2 + (54-50)^2 + (8-8)^2}$

$= 4.47$

② (27, 54, 8) to (45, 80, 3)

$= \sqrt{(27-45)^2 + (54-80)^2 + (8-3)^2}$

$= 32.01$

③ (27, 54, 8) to (28, 55, 9)

$= \sqrt{(27-28)^2 + (54-54)^2 + (8-9)^2}$

$= 1.713$

④ (27, 54, 8) to (50, 85, 2)

$= \sqrt{(27-50)^2 + (54-85)^2 + (8-2)^2}$

$= 39.06$

⑤ (27, 54, 8) to (30, 60, 10)

$= \sqrt{(27-30)^2 + (54-60)^2 + (8-10)^2}$

$= 7$

⑥ (27, 54, 8) to (48, 82, 3)

$= \sqrt{(27-48)^2 + (54-82)^2 + (8-3)^2}$

$= 35.35$

(7) (27, 54, 8) to (26, 52, 8)

$= \sqrt{(27-26)^2 + (54-52)^2 + (8-8)^2}$

$= 2.236$

(8) (27, 54, 8) to (47, 88, 2)

$= \sqrt{(27-47)^2 + (54-88)^2 + (8-2)^2}$

$= 39.89$

(9) (27, 54, 8) to (29, 58, 9)

$= \sqrt{(27-29)^2 + (54-58)^2 + (8-9)^2}$

$= 4.58$

9 training samples $\boxed{k=5}$

4.47 y

32.01

1.713 y

39.06

7 y

35.35

2.236 y

39.89

4.58 y

Distance for (49, 86, 2) to (25, 50, 8)

(45, 80, 3)

(28, 55, 9)

(50, 85, 2)

(30, 60, 10)

(48, 82, 3)

(26, 52, 8)

(47, 88, 2)

(29, 58, 9)

① (49, 86, 2) to (25, 50, 8)

$= \sqrt{(49-25)^2 + (86-50)^2 + (2-8)^2}$

$= 43.68$

② (49, 86, 2) to (45, 80, 3)

$= \sqrt{(49-45)^2 + (86-80)^2 + (2-3)^2}$

$= 7.28$

③ (49, 86, 2) to (28, 55, 9)

$= \sqrt{(49-28)^2 + (86-55)^2 + (2-9)^2}$

$= 38.09$

④ (49, 86, 2) to (50, 85, 2)

$= \sqrt{(49-50)^2 + (86-85)^2 + (2-2)^2}$

$= 1.41$

⑤ (49, 86, 2) to (30, 60, 10)

$= \sqrt{(49-30)^2 + (86-60)^2 + (2-10)^2}$

$= 33.18$

⑥ (49, 86, 2) to (48, 82, 3)

$= \sqrt{(49-48)^2 + (86-82)^2 + (2-3)^2}$

$= 4.24.$

(7) (49, 86, 2) to (26, 52, 8)

$= \sqrt{(49-26)^2 + (86-52)^2 + (2-8)^2}$

$= 41.48$

(8) (49, 86, 2) to (47, 88, 2)

$= \sqrt{(49-47)^2 + (86-88)^2 + (2-2)}$

$= 2.82$

(9) (49, 86, 2) to (29, 58, 9)

$= \sqrt{(49-29)^2 + (86-58)^2 + (2-9)^2}$

$= 35.11$

9 training samples

43.68

7.28 N

38.09

1.41 N

33.18 N

4.24 N

41.48

2.82 N

35.11